

QR Decomposition

- Used to solve Non-square / Overdetermined matrix.
- Modified version of Least Square Approximation.

Here

$$A = QR$$

coefficient matrix
orthonormal matrix
calculated upper triangular matrix

We know

$$A^T A X = A^T B$$

$$\Rightarrow (QR)^T A X = (QR)^T B$$

$$\Rightarrow R^T \underbrace{Q^T Q}_=1 R X = R^T Q^T B$$

$$\Rightarrow R^T R X = R^T Q^T B$$

Final
Formula

$$\boxed{\Rightarrow R X = Q^T B}$$

[R^T both side removed]

To calculate $Q \rightarrow$ we need

Gram-Schmidt
method

Gram Schmidt method

$$A = \begin{bmatrix} 3 & 1 \\ 6 & 2 \\ 0 & 2 \end{bmatrix}$$

$\downarrow \quad \downarrow$
 $u_1 \quad u_2$

must give
a scalar
value!

Now, using GS method: Formula $P_k = U_k - \sum_{i=1}^{k-1} (U_k^T q_i) q_i$

$$q_k = \frac{P_k}{|P_k|}$$

Step-1
 $K=1$

$$\checkmark P_1 = u_1 - 0 = \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

$$\checkmark q_1 = \frac{P_1}{|P_1|} = \frac{1}{\sqrt{3^2+6^2+0^2}} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{45}} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

Step 2 $K=2$

$$P_2 = u_2 - (u_2^T q_1) q_1$$

$$= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \left((1 \ 2 \ 2) \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix} \frac{1}{\sqrt{45}} \right) q_1$$

$$= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \left(15 \times \frac{1}{\sqrt{45}} \right) \left(\frac{1}{\sqrt{45}} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix} \right)$$

$$= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

$$q_2 = \frac{p_2}{|p_2|} = \frac{1}{\sqrt{2^2+0^2+0^2}} \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$Q = (q_1 \quad q_2) = \begin{pmatrix} \frac{3}{\sqrt{45}} & 0 \\ 6/\sqrt{45} & 0 \\ 0 & 1 \end{pmatrix}$$

that's GS method.

Let's do an Example

Given

$f(-3)=0$
 $f(0)=0$
 $f(6)=2$

need overdetermined matrix.
So, degree (1).

$$P_1 = a_0 + a_1 x = f(x)$$

equations:

$$\begin{cases} a_0 + (-3)a_1 = 0 \\ a_0 + 0a_1 = 0 \\ a_0 + 6a_1 = 2 \end{cases}$$

$$A = \begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix}; \quad x = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}; \quad B = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$\downarrow \quad \downarrow$
 $u_1 \quad u_2$

Step 1:

$K=1$

$$\checkmark p_1 = u_1 - 0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\checkmark q_1 = \frac{p_1}{|p_1|} = \frac{1}{\sqrt{1^2+1^2+1^2}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Step 2

$K=2$

$$\checkmark p_2 = u_2 - (u_2^T q_1) q_1$$

$$= \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} - \left((-3 \ 0 \ 6) \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} \right) q_1$$

$$= \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} - \sqrt{3} \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$$

$$= \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\rightarrow = \begin{pmatrix} -4 \\ -1 \\ 5 \end{pmatrix}$$

$$\checkmark q_2 = \frac{p_2}{|p_2|} = \frac{1}{\sqrt{(-4)^2+(-1)^2+5^2}} \begin{pmatrix} -4 \\ -1 \\ 5 \end{pmatrix} = \frac{1}{\sqrt{42}} \begin{pmatrix} -4 \\ -1 \\ 5 \end{pmatrix}$$

$$Q = (q_1 \quad q_2)$$

$$\therefore Q = \begin{pmatrix} \frac{1}{\sqrt{3}} & -4/\sqrt{42} \\ 1/\sqrt{3} & -1/\sqrt{42} \\ 1/\sqrt{3} & 5/\sqrt{42} \end{pmatrix}$$

we got $Q!!$

What will be R??

$$\text{Size of } R = |u| \times |u|$$

$|u|$ = number of u matrix we used!

$$\text{if } R_{2 \times 2} = \begin{bmatrix} u_1^T q_1 & u_2^T q_1 \\ 0 & u_2^T q_2 \end{bmatrix}; \quad \text{if } R_{3 \times 3} = \begin{bmatrix} u_1^T q_1 & u_2^T q_1 & u_3^T q_1 \\ 0 & u_2^T q_2 & u_3^T q_2 \\ 0 & 0 & u_3^T q_3 \end{bmatrix}$$

in our example, we'd $\overbrace{u_1, u_2}^{u}$. So, R will be 2×2 .
 $|u| = 2$

$$R = \begin{bmatrix} u_1^T q_1 & u_2^T q_1 \\ 0 & u_2^T q_2 \end{bmatrix}$$

$\rightarrow u_1^T q_1 = (1 \ 1 \ 1) \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} = \sqrt{3}$

$\rightarrow u_2^T q_1 = (-3 \ 0 \ 6) \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} = \sqrt{3}$

$\rightarrow u_2^T q_2 = (-3 \ 0 \ 6) \begin{pmatrix} -4/\sqrt{42} \\ -1/\sqrt{42} \\ 5/\sqrt{42} \end{pmatrix} = \sqrt{42}$

$$\therefore R = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix}$$

Now $RX = Q^T B$

$$\Rightarrow \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{3} \\ 10/\sqrt{42} \end{bmatrix}$$

$$\left. \begin{aligned} \therefore \sqrt{42} a_1 &= 10/\sqrt{42} \Rightarrow a_1 = 5/21 \\ \therefore \sqrt{3} a_0 + \sqrt{3} a_1 &= 2/\sqrt{3} \Rightarrow a_0 = 3/7 \end{aligned} \right\} \underline{\underline{\text{ANS.}}}$$

Steps

0. Define $u_1, u_2, \dots$

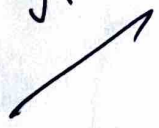
1. Find P_s, q_s

2. Build $Q = (q_1 \ q_2 \ q_3 \dots)$

3. Build R matrix

4. Use $RX = Q^T B$.

5. Find X matrix by solving.



Ex-2

$$\begin{aligned} f(1) &= 2 \\ f(2) &= 3 \\ f(3) &= 6 \\ f(4) &= 4 \end{aligned} \quad \left\{ \begin{aligned} P_2(x) &= a_0 + a_1x + a_2x^2 \\ \Rightarrow P_2(1) &\Rightarrow a_0 + a_1 + a_2 = 2 \\ \Rightarrow P_2(2) &\Rightarrow a_0 + 2a_1 + 4a_2 = 3 \\ \Rightarrow P_2(3) &\Rightarrow a_0 + 3a_1 + 9a_2 = 6 \\ \Rightarrow P_2(4) &\Rightarrow a_0 + 4a_1 + 16a_2 = 4 \end{aligned} \right.$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} ; X = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} ; B = \begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix}$$

$u_1 \quad u_2 \quad u_3$

Step 1 : $K=1$

$$\checkmark P_1 = u_1 - 0 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\checkmark q_1 = \frac{P_1}{|P_1|} = \frac{1}{\sqrt{1^2+1^2+1^2+1^2}} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{pmatrix}$$

Step 2 : $K=2$

$$\checkmark P_2 = u_2 - (u_2^T q_1) q_1$$

$$= \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} - \left((1 \ 2 \ 3 \ 4) \begin{pmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{pmatrix} \right) \cdot \begin{pmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{pmatrix}$$

$$\rightarrow = \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix}$$

$$\checkmark q_2 = \frac{P_2}{|P_2|} = \frac{1}{\sqrt{(-1.5)^2 + (-0.5)^2 + 0.5^2 + 1.5^2}} \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix}$$

Step-3 $K=3$

$$P_3 = U_3 - \left\{ (U_3^T q_1) q_1 + (U_3^T q_2) q_2 \right\}$$

$$= \begin{pmatrix} 1 \\ 4 \\ 9 \\ 16 \end{pmatrix} - \left\{ \left((1 \ 4 \ 9 \ 16) \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} \right) \cdot \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} + (1 \ 4 \ 9 \ 16) \begin{pmatrix} -1.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ 0.5/\sqrt{5} \\ 1.5/\sqrt{5} \end{pmatrix} \right\}$$

$$= \begin{pmatrix} 1 \\ 4 \\ 9 \\ 16 \end{pmatrix} - \left\{ 15 \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} + 5\sqrt{5} \begin{pmatrix} -1.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ 0.5/\sqrt{5} \\ 1.5/\sqrt{5} \end{pmatrix} \right\}$$

$$= \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

$$q_3 = \frac{P_3}{|P_3|} = \frac{1}{\sqrt{1^2 + (-1)^2 + (-1)^2 + 1^2}} \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.5 \\ -0.5 \\ -0.5 \\ 0.5 \end{pmatrix}$$

Do this calculation patiently!
use: Calculator!

$$Q = (q_1 \quad q_2 \quad q_3)$$

$$= \begin{pmatrix} 0.5 & -1.5/\sqrt{5} & 0.5 \\ 0.5 & -0.5/\sqrt{5} & -0.5 \\ 0.5 & 0.5/\sqrt{5} & -0.5 \\ 0.5 & 1.5/\sqrt{5} & 0.5 \end{pmatrix}$$

$$R = \begin{bmatrix} U_1^T q_1 & U_2^T q_1 & U_3^T q_1 \\ 0 & U_2^T q_2 & U_3^T q_2 \\ 0 & 0 & U_3^T q_3 \end{bmatrix}$$

$$U_3^T q_1 = (1 \ 4 \ 9 \ 16) \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} = 15$$

$$U_3^T q_3 = (1 \ 4 \ 9 \ 16) \begin{pmatrix} 0.5 \\ -0.5 \\ -0.5 \\ 0.5 \end{pmatrix} = 2$$

$$= \begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix}$$

Similarly others!

Finally

$$RX = Q^T B$$

$$\Rightarrow \begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ -1.5/\sqrt{5} & 0.5/\sqrt{5} & 0.5/\sqrt{5} & 1.5/\sqrt{5} \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 5.5 \\ 27\sqrt{5}/10 \\ -3.5 \end{bmatrix}$$

use bottom to top to find,

$$\left. \begin{array}{l} a_0 = ? \\ a_1 = ? \\ a_2 = ? \end{array} \right\} \begin{array}{l} \text{same as} \\ \text{before!} \end{array}$$